

- An existing device that is conflicting with the new device may suffer performance problems
- Windows starts in Safe Mode
- The system crashes and you get a BSOD (Blue Screen of Death)
- The sound card stops working or stutters
- The screen jumps and shows garbled images
- The mouse stops working
- Modem/network card doesn't respond
- CD/DVD drive doesn't respond or crashes the system

1.5.2 Problems causing conflicts

A wide range of problems may be causing the above and other symptoms. Assuming that the hardware is not defective and has been installed correctly, at the operating system level the problems can be broadly classified as follows:

- There could be a resource allocation (IRQ/memory address) conflict. Two devices are allocated the same IRQ/memory address space. Usually, non-Plug and Play devices would give this problem. Since Windows is unable to detect these devices it may not allocate the resources properly and hence cause conflicts with an existing device.

Note: Some devices, such as PCI devices, can share a single IRQ without conflicts in Windows 2000/XP.

Concepts

IRQ

Interrupt ReQuest lines. These are communication lines over which all hardware components talk to the processor.

DMA

Direct Memory Access enables hardware devices to transfer data directly from memory (RAM) without going through the processor.

I/O Ports

Input/Output ports: the connection points between the motherboard and hardware devices.

Memory Address

A specific physical location in the RAM chip.

Windows Registry

A database that stores all the configuration information of the computer.